Capstone: five builds from first principles, from linear regression to attention

Can five small builds, written from first principles, reproduce the core of what runs inside modern AI?

Mathematics for AI and Machine Learning | Session 18 of 18 | Capstone

KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Build linear regression (normal equation and gradient descent) and logistic regression (gradient descent) with numpy

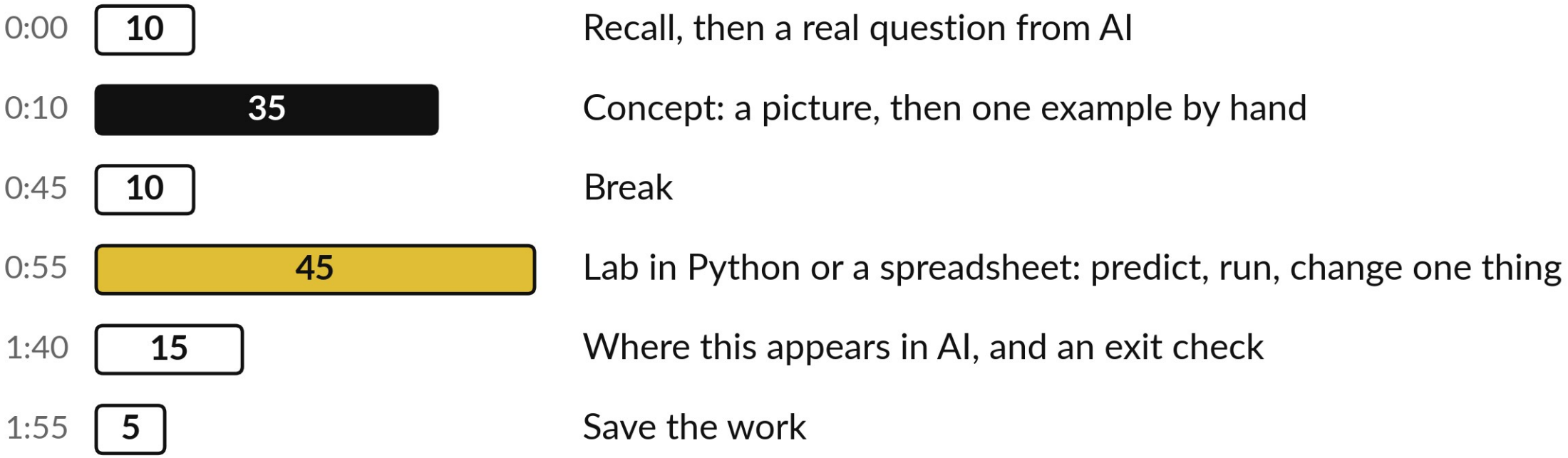


Build PCA from the covariance eigen-decomposition and a 2-4-1 neural network with backpropagation



Build scaled dot-product attention for three tokens, and pass the automatic checks of the capstone

Your two hours: a guided tour, then the capstone



The tour first. Then the individual capstone, graded in the notebook. Then AI.

Three questions from last time

1. How are cross-entropy, entropy and KL divergence related?

2. What is the gradient of softmax with cross-entropy with respect to the logits?

3. What does a perplexity of 5 mean?



Answer aloud first. Then we check together.

Answers to the three questions from Session 17

Question	Short answer
How are cross-entropy, entropy and KL divergence related?	Cross-entropy = entropy + KL divergence. The entropy is fixed by the data, so lowering the cross-entropy lowers the KL divergence.
What is the gradient of softmax with cross-entropy with respect to the logits?	The prediction minus the target: p - y.
What does a perplexity of 5 mean?	The model is as unsure as if it chose among about five equally likely words at each step.

All three come back today: p - y trains the network, and the softmax is inside attention.

Words we use today

	Word	What it means
	**Normal equation**	The direct formula w = (X^T X)^-1 X^T y that gives the least squares weights.
	**Principal component**	A direction of largest spread in the data: an eigenvector of the covariance matrix.
	**Backpropagation**	The chain rule applied backwards through a network to get the gradient of the loss for every weight.
	**Query, key, value**	The three vectors each token makes in attention: what it looks for, what it offers and what it passes on.
	**Attention**	Each token takes a weighted average of the value vectors, with weights from softmax(Q K^T / sqrt(d)).

Five builds, and each one uses mathematics you already know



Linear regression

Linear systems and least squares



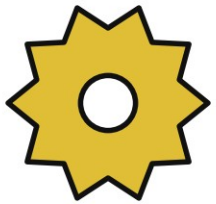
Logistic regression

Gradient descent and cross-entropy



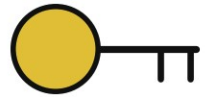
PCA

Covariance, eigenvalues, eigenvectors



Neural network

Chain rule and backpropagation



Attention

Matrices, dot products and softmax

Illustrative data. Each build is about ten lines of numpy.

The illustrative data of Builds 1 to 3 (made-up numbers to show the arithmetic)

Build	Data
1 Linear	Hours x = 1, 2, 3, 4, 5, 6. Energy y (kWh) = 2.2, 2.9, 4.1, 5.2, 5.8, 7.1.
2 Logistic	Hours x = 1 to 8. Passed y = 0, 0, 0, 1, 0, 1, 1, 1.
3 PCA	a = 2.5, 0.5, 2.2, 1.9, 3.1, 2.3, 2.0, 1.0, 1.5, 1.1. b = 2.4, 0.7, 2.9, 2.2, 3.0, 2.7, 1.6, 1.1, 1.6, 0.9.

These fixed numbers are for the slides and the guided workbook. In the graded notebook your own student ID generates different data.

The illustrative data of Builds 4 and 5

Build	Data
4 Network	W1 rows (0.5, -0.3), (0.2, 0.8), (-0.6, 0.4), (0.1, -0.9). b1 = 0. W2 = (0.3, -0.5, 0.7, 0.2). b2 = 0.1.
4 XOR rows	Inputs (0, 0), (0, 1), (1, 0), (1, 1) with targets 0, 1, 1, 0.
***5* Attention**	X rows (1, 0), (0, 1), (1, 1). Wq rows (1, 0), (1, 1). Wk rows (1, 1), (0, 1). Wv rows (1, 0), (0, 2). d = 2.

Q = X Wq, K = X Wk and V = X Wv give three rows of two numbers each.

Build 1: linear regression has a direct solution and an iterative one

Normal equation

$$w = (X^T X)^{-1} X^T y$$

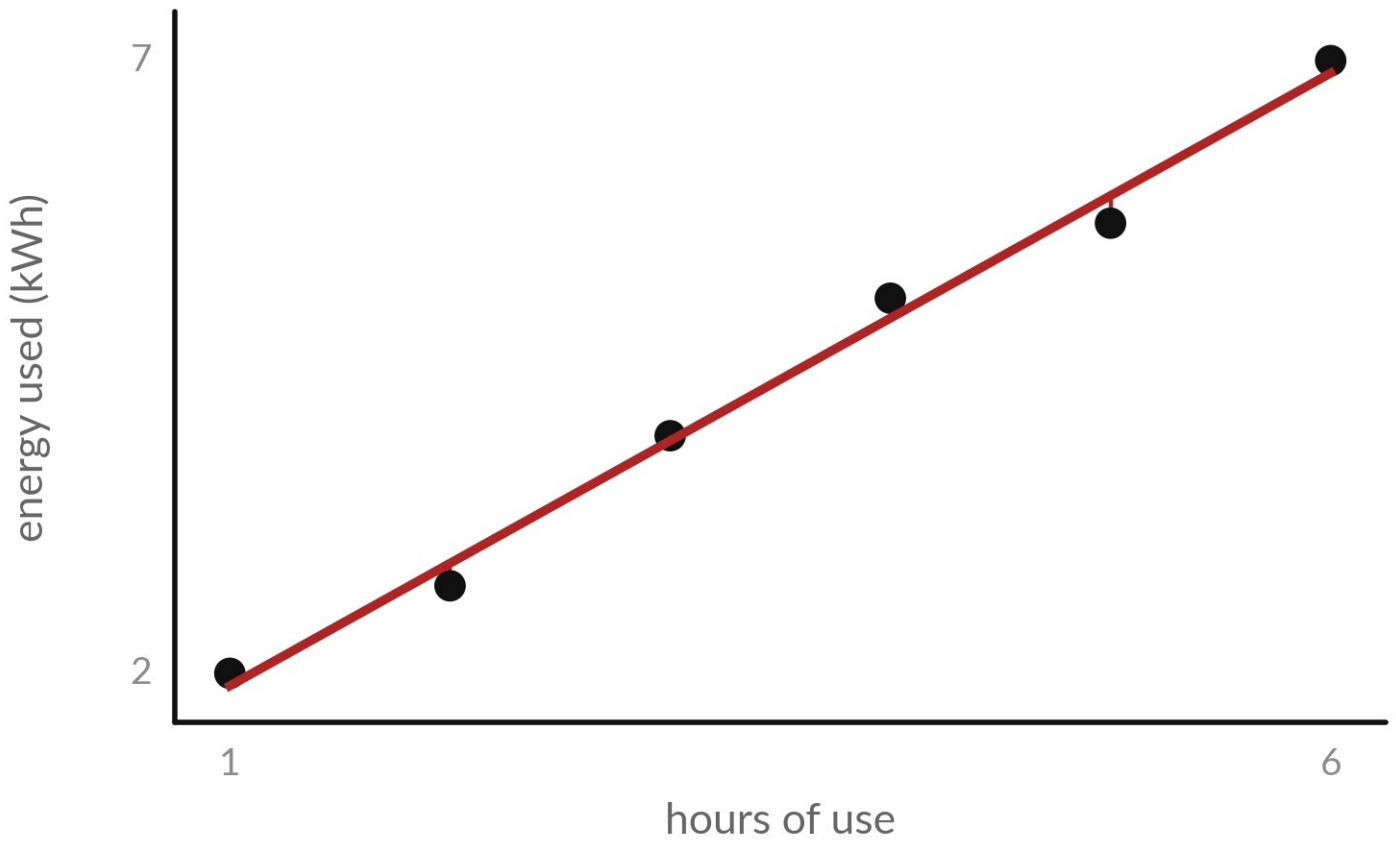
Gradient descent step

$$w \leftarrow w - \eta \frac{2}{n} X^T (Xw - y)$$

Symbol	What it means	In our data
X	The design matrix: a column of ones, then x.	6 rows by 2 columns
w	The weights: intercept w0 and slope w1.	(1.12, 0.98)
eta	The learning rate.	0.02
n	The number of rows.	6

Same weights from both. MSE 0.0213. 7 hours (beyond the data) predicts 7.98 kWh.

Build 1 in a picture: the fitted line and its residuals



The line is y-hat = 1.12 + 0.98 x. The red segments are the residuals.

Illustrative data: x = 1 to 6 hours, y = 2.2, 2.9, 4.1, 5.2, 5.8, 7.1 kWh.

The normal equation and gradient descent give the same line.

Build 2: logistic regression is a sigmoid on a linear score, trained by gradient descent

Model

$$p = \sigma(w_0 + w_1 x) = \frac{1}{1 + e^{-(w_0 + w_1 x)}}$$

Gradient descent step

$$w \leftarrow w - \eta \frac{1}{n} X^T (p - y)$$

Symbol	What it means	In our data
p	The predicted probability of class 1 (passed).	0.5 at the start
p - y	The prediction minus the target, as in Session 17.	one value per row
eta	The learning rate.	0.1
Loss	Binary cross-entropy, from Session 16.	0.693 at the start

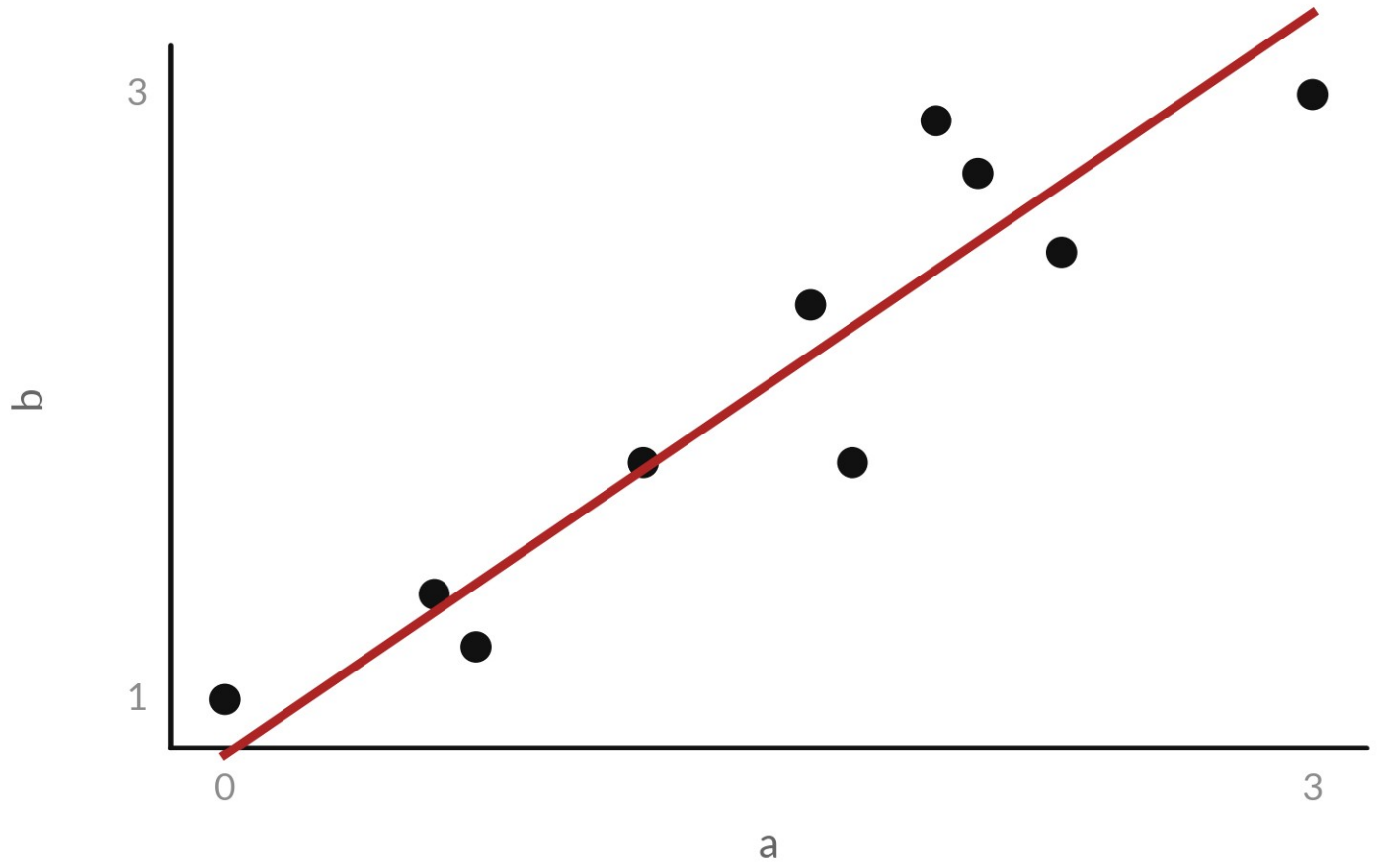
After 5 steps: w = (-0.0509, 0.1494), loss 0.623. Data: 8 students (see the data slide).

Build 2 by hand: the loss falls at every one of the first five gradient steps

Step	w0	w1	Loss
0	0.0000	0.0000	0.6931
1	0.0000	0.0875	0.6407
2	-0.0096	0.1207	0.6316
3	-0.0224	0.1359	0.6281
4	-0.0364	0.1442	0.6255
5	-0.0509	0.1494	0.6232

Learning rate 0.1. Step 0 is the start. The workbook has the same table.

Build 3: PCA finds the direction of largest spread in the data



The red line is the first principal component. It holds 96.3 percent of the variance.

Eigenvalues of the covariance matrix: 1.156 and 0.044.

First component: (0.678, 0.735).

PCA as formulas: centre, form the covariance, find its eigenvectors, project

Covariance matrix (divide by n)

$$C = \frac{1}{n} X_c^T X_c$$

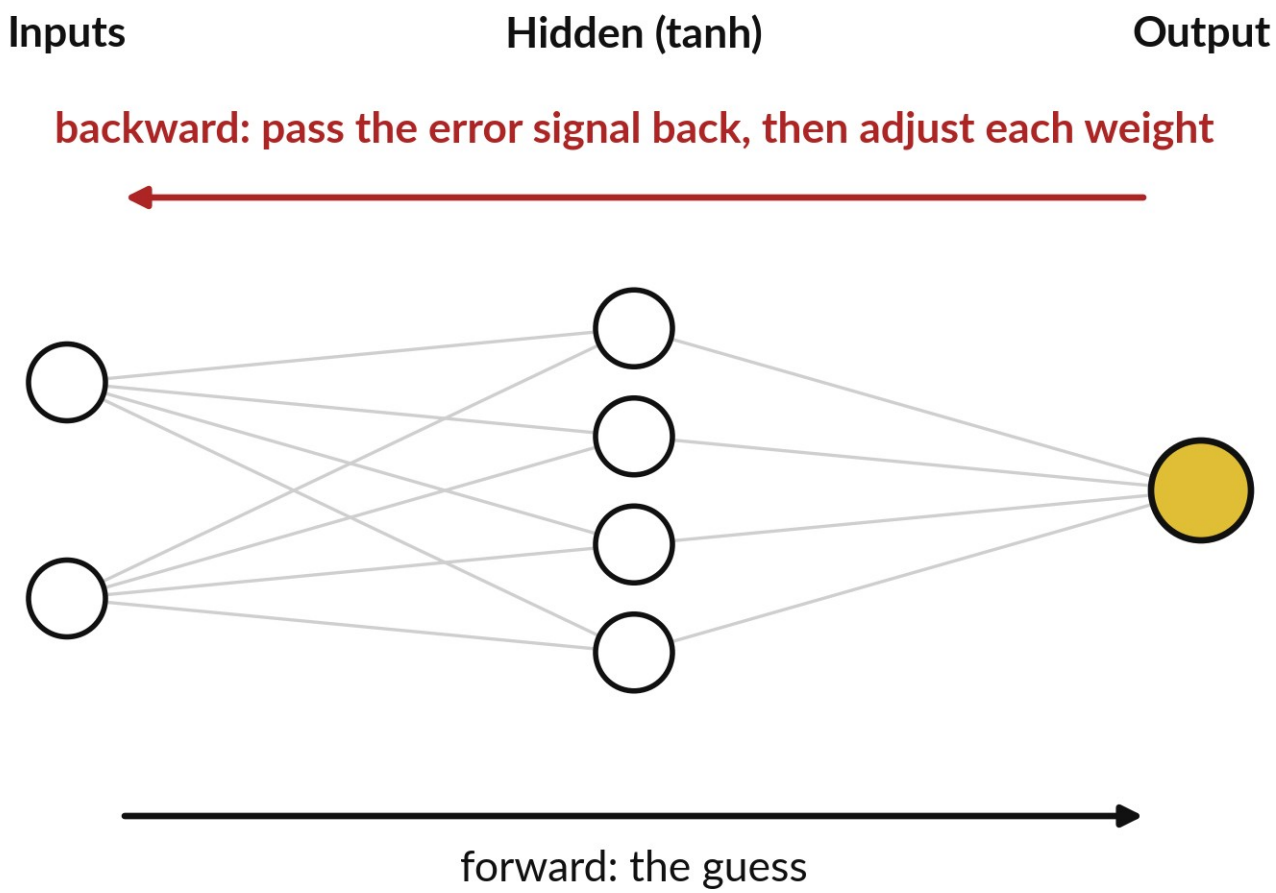
Eigenvector and score

$$Cv = \lambda v, \qquad \text{score} = X_c\, v_1$$

Symbol	What it means	In our data
X_c	The data with each column centred on its mean.	10 rows by 2 columns
lambda	An eigenvalue: the variance along its eigenvector.	1.156 and 0.044
v_1	The eigenvector with the largest eigenvalue: the first component.	(0.678, 0.735)
n	The number of points.	10

Libraries divide by n - 1: eigenvalues 1.284 and 0.049, the same 96.3 percent share.

Build 4: a 2-4-1 network has two layers of weights, and the error flows back



Two inputs, four hidden units, one output. The task is XOR: 1 when exactly one input is 1.

Forward: h = tanh(W1 x + b1), then p = sigmoid(W2 h + b2). Loss: binary cross-entropy.

Backward: the chain rule passes the error signal back, layer by layer.

Backpropagation for the 2-4-1 network, in matrix algebra

Forward and loss $$h = \tanh(W_1 x + b_1), \quad p = \sigma(W_2 h + b_2), \quad L = -[\, y \ln p + (1-y)\ln(1-p)\,]$$

Backward $$\delta = p - y, \quad g_{W_2} = \delta\, h^T, \quad g_{b_2} = \delta, \quad \delta_h = (W_2^T \delta) \odot (1 - h^2), \quad g_{W_1} = \delta_h\, x^T, \quad g_{b_1} = \delta_h$$

Symbol	What it means	In our data
delta	The error signal at the output. For a sigmoid with binary cross-entropy it is p - y.	-0.6112 in the example
delta_h	The error signal arriving at each hidden unit.	one value per unit
1 - h^2	The slope of tanh at each hidden unit (chain rule).	(0.990, 0.104, 0.961, 0.125)
odot	Multiply entry by entry.	unit by unit

Update with learning rate eta: each weight matrix and bias becomes W - eta × its gradient.

By hand: one forward and one backward step for the input (1, 2) with target 1

One illustrative example (not an XOR row), from the starting weights on the data slide. The notebook trains on many rows and averages their gradients.

Forward	h = (-0.100, 0.947, 0.197, -0.935), z2 = -0.4522, p = 0.3888
Loss	-ln p = 0.9446
Backward	delta = p - y = -0.6112. g_W2 = delta × h = (0.061, -0.579, -0.121, 0.572). First row of g_W1 = (-0.182, -0.363)
Update	W2 first weight: 0.3 - 0.5 × (0.061) = 0.270

After 5,000 gradient steps the toy network solves XOR (guided example)

Input	Target	Output before training	Output after training
(0, 0)	0	0.525	0.0002
(0, 1)	1	0.451	0.9989
(1, 0)	1	0.446	0.9992
(1, 1)	0	0.379	0.0011

The loss falls from 0.706 to 0.0008. Learning rate 0.5.

Build 5: attention gives every token a weighted average of the value vectors

rows = query tokens (asking) columns = key tokens (being read)

	Token 1	Token 2	Token 3	output = weights × V
Token 1	**0.401**	0.198	**0.401**	→ (0.80, 1.20)
Token 2	0.284	0.140	**0.576**	→ (0.86, 1.43)
Token 3	0.306	0.074	**0.620**	→ (0.93, 1.39)

Illustrative numbers. Token 3 reads 31 percent from token 1, 7 from token 2 and 62 from token 3.

Each row of weights adds to 1. The output is the weights times the value vectors.

Attention as a formula: every symbol named

Projections

$$Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V$$

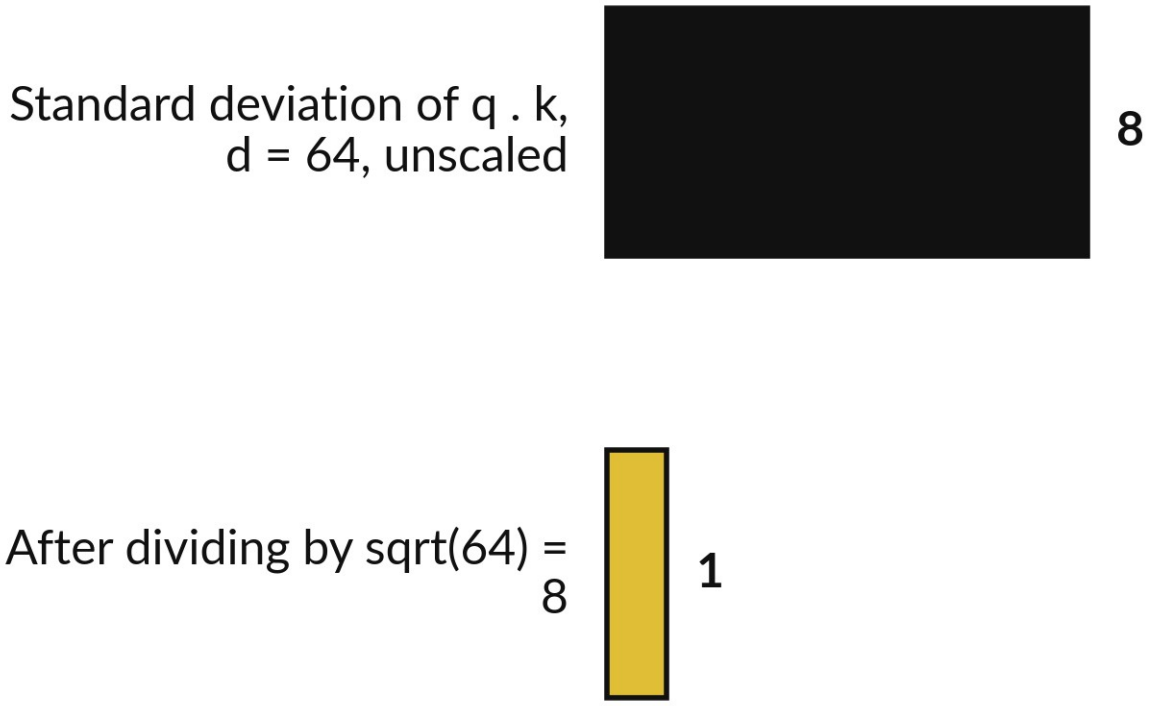
Attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Symbol	What it means	In our data
X	One row per token: its embedding.	3 tokens, 2 numbers each
W_Q, W_K, W_V	Weight matrices that make queries, keys and values.	learned in a real model
Q, K, V	Queries (what a token looks for), keys (what it offers), values (what it passes on).	each 3 by 2
d_k	The length of a key vector.	2
softmax	Applied along each row, so each row adds to 1.	weights 0.401, 0.198, 0.401

Vaswani et al. (2017): large dot products saturate the softmax, so its gradient is tiny.

Why divide by the square root of d: dot products of long vectors grow, and the softmax saturates



If the entries of q and k are independent with mean 0 and standard deviation 1, then q . k has mean 0 and standard deviation sqrt(d).

Large scores make the softmax saturate: almost all the weight goes to one token (nearly one-hot), and the gradient is tiny.

Dividing by sqrt(d_k) keeps the standard deviation near 1. The notebook simulates 2,000 pairs.

By hand: scaled scores, weights and output for three tokens (d = 2)

Token	Scaled scores	Weights	Output
Token 1	0.707, 0.000, 0.707	0.401, 0.198, 0.401	0.802, 1.198
Token 2	1.414, 0.707, 2.121	0.284, 0.140, 0.576	0.860, 1.432
Token 3	2.121, 0.707, 2.828	0.306, 0.074, 0.620	0.926, 1.389

Illustrative. Scores = Q K^T / sqrt(2). Weights = softmax of each row. Output = weights times V.

Three slips that the capstone checks will catch

The slip	The fix
No column of ones, so no intercept.	Add the column of ones to X.
PCA without centring the data first.	Subtract each column mean first.
Softmax along the wrong axis, or no sqrt(d).	Softmax along each row, then divide scores by sqrt(d_k).

Each check names its build, so read the name first.

Read the five recipes before you write them

```
w = np.linalg.solve(X.T @ X, X.T @ y)
w = w - lr * X.T @ (sigmoid(X @ w) - y) / n
vals, vecs = np.linalg.eigh(Dc.T @ Dc / n)
gW1 = dh.T @ X
out = softmax_rows(Q @ K.T / np.sqrt(d)) @ V
```

In plain words

Build 1: the normal equation in one line.

Build 2: one gradient step of logistic regression.

Build 3: eigh returns the eigenvalues smallest first, so take the last column. The sign of an eigenvector is arbitrary.

Build 4: a gradient from backpropagation.

Build 5: the attention formula.

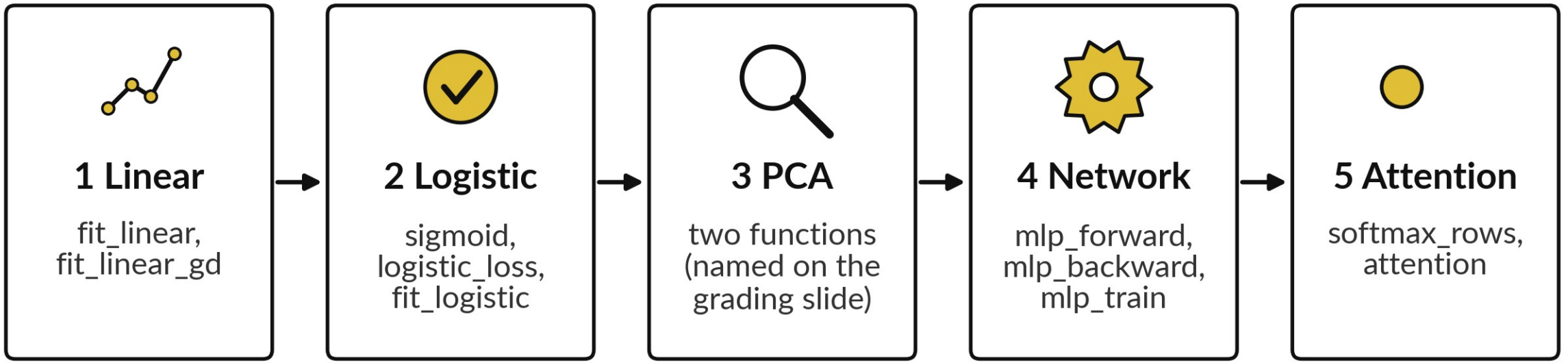
Spreadsheet: MMULT, MINVERSE, TRANSPOSE and SUMPRODUCT.



Part 2: the capstone

Close the slides. Open the notebook or the workbook.

The capstone has five builds, and every check names its build



Python: 41 checks, each run on your own data and on a second dataset. 20 marks per build. Spreadsheet: a guided version with fixed toy numbers.

How the capstone is graded: your own data and your own functions

	Step	What happens
	**1 Student ID**	Type it in the first cell. Every dataset is generated from it, so your numbers are yours alone.
	**2 Functions**	You write 12 functions with fixed names: fit_linear, fit_linear_gd, sigmoid, logistic_loss, fit_logistic, pca_first_component, pca_variance_share, mlp_forward, mlp_backward, mlp_train, softmax_rows, attention.
	**3 Checks**	Each check runs your function on your data and on a second dataset, and compares with a trusted reference. No expected number is shown anywhere.
	**4 Marking**	Your teacher runs your functions again on datasets you cannot see. Typed-in numbers fail. 20 marks per build, 100 in total.

The spreadsheet is a guided practice version with fixed toy numbers. It is not the graded capstone.

Guided workbook: check your numbers against these (toy data)

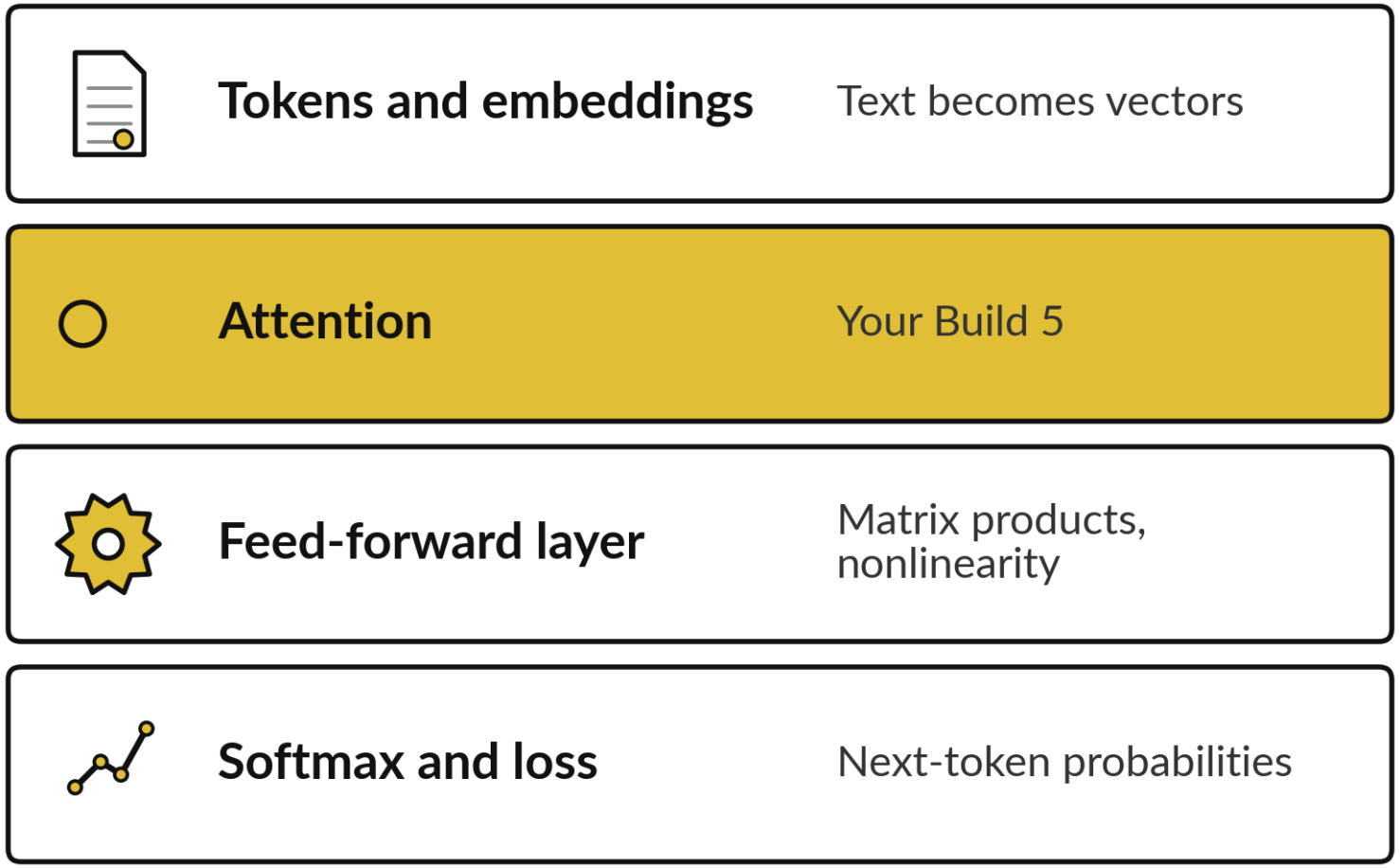
	Build	What you should see
	**1 Linear regression**	weights 1.12, 0.98; MSE 0.0213; 7 hours gives 7.98
	**2 Logistic regression**	after 5 steps w = (-0.0509, 0.1494), loss 0.6232
	**3 PCA**	eigenvalues 1.1556, 0.0442 (largest first); first component (0.6779, 0.7352); share 0.9632
	**4 Neural network**	input (1, 2), target 1: p = 0.3888, loss 0.9446, delta = -0.6112, first g_W2 = 0.0609
	**5 Attention**	weights row 1: 0.401, 0.198, 0.401; output row 3: 0.926, 1.389

These toy numbers are for the guided workbook. The graded notebook shows no expected numbers. In numpy, eigh gives the eigenvalues smallest first.

Errors you will meet, what they mean and what to do

	What you see	What it means	What to do
	**TypeError: NoneType**	A function still says return None	Write your code in place of the placeholder
	**ValueError: shapes not aligned**	A matrix product has the wrong shapes	Print the shapes. Transpose one side
	**A check says NOT YET**	Your function disagrees with the reference on your data or on the second dataset	Call it on a tiny example and work it out by hand
	**Spreadsheet shows #VALUE!**	An array formula was not filled over the whole gold block	Select the block and enter it as an array formula

Where this appears in AI: attention is the core step of every layer of a published transformer



Published research describes transformer language models as many repeated layers of attention and feed-forward steps.

Your Build 5 is the published attention formula (Vaswani et al., 2017).

Closed products do not publish their designs. Open models such as Llama use the same core formula plus a causal mask, and other changes.

Where this appears in AI: the five builds in published models

	Your build	Where the same idea appears
	**Linear regression**	Every layer multiplies by a weight matrix: a linear map. Its weights are learned by gradient descent, a different method from least squares.
	**Logistic regression**	The last step of a classifier is a softmax of linear scores (softmax regression, the many-class version), trained with cross-entropy.
	**PCA**	Used to inspect and compress embeddings. LoRA (Hu et al., 2021) learns a low-rank update to a weight matrix: the same low-rank idea that makes the first principal components useful.
	**Neural network**	Backpropagation runs in every training step of published models.
	**Attention**	The core step of the transformer, repeated in every layer.

Phrased as published research describes. No claim about the internals of any closed product.

Check an AI assistant's attention against your own numbers

5 minutes Optional, with any assistant

Give an assistant the three-token example: X, Wq, Wk, Wv from the data slide, d = 2. Ask for the attention weights.

Compare its row 1 with 0.401, 0.198, 0.401.

Ask if it divided by the square root of d. Try without, and see which weights get sharper.

Never paste confidential data into an assistant.

You keep

A note: where the assistant agreed with your numbers, and where it did not.

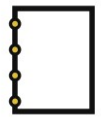
What to learn next: four steps



Machine Learning at KLAi: use these builds on real data, with scikit-learn.



A deep learning course, where PyTorch automatic differentiation replaces your hand-written backpropagation (title and dates to be confirmed by KLAi).



Read section 3.2 of Vaswani et al. (2017), "Attention Is All You Need", and match every symbol to your Build 5.



Rebuild the five builds from a blank notebook, then write your one-page formula sheet.

Exit check: three questions and their short answers

Question	Short answer
What does each row of the attention weights add up to, and why?	To 1, because a softmax is applied along each row.
Why is the attention score divided by the square root of d_k?	Large dot products make the softmax saturate: almost all the weight goes to one token and the gradient is tiny. The scaling keeps the standard deviation near 1.
In PCA, which eigenvector is the first principal component?	The one with the largest eigenvalue, the direction of largest variance.

Answer each in one sentence before you read the right-hand column.

Save your capstone and write your formula sheet

What you keep from today

Keep your finished capstone notebook with all checks passing, and write your one-page formula sheet: every formula named in plain words and in the industry term.

Our question today

Can five small builds, written from first principles, reproduce the core of what runs inside modern AI?

What to learn next

Machine Learning at KLAi, then deep learning (to be confirmed).

Take away

Capstone
All checks pass

Formula sheet
One page

Next
Machine Learning at KLAi

Then
Deep learning